

DISASTER & CRISIS RESPONSE HACKATHON

PRATHAMIKTA

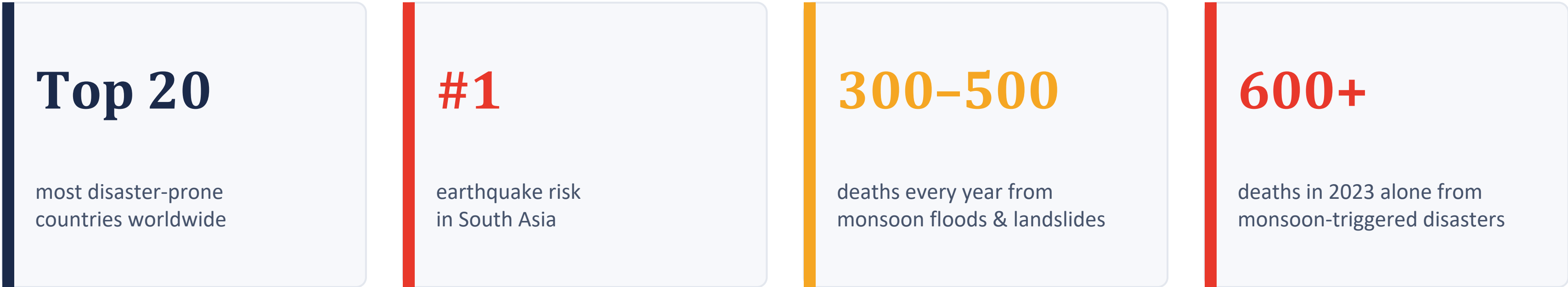
प्राथमिकता

Team LumneX



Nepal's Disaster Reality

Nepal ranks among the top 20 most disaster-prone countries in the world — yet response still runs on phone calls, WhatsApp, and spreadsheets.



2015 Gorkha Earthquake — 9,000+ lives lost across 77 districts and 7 provinces, geographically fragmented and remote, exposing the need for a coordinated national response system.

The Problem

1

Information Chaos

Reports pour in from 100call centers, GDACS sensor alerts, social media, and news RSS feeds — with no unified system to collect and merge them.

2

No Priority Ranking

Dispatchers manually decide which disaster gets the first rescue team — subjective, inconsistent, and slow under pressure.

3

Duplicate Reports Flood the Queue

One landslide can generate 10 calls, 3 articles, and 2 sensor alerts — all treated as separate incidents.

4

No Real-Time Situational Awareness

No live map showing active disasters, rescue status, and resource proximity at a glance.

***“Not all disasters are equal.
The system must score them —
not the dispatcher.”***

1

Classified

by disaster type
and severity

2

Fused

with related reports into
one consolidated incident

3

Scored

using a transparent,
weighted formula

4

Ranked

so dispatchers see the most
critical disaster first

PRATHAMIKTA: Features

Nepali: प्राथमिकता, meaning "Priority" — a real-time, AI-powered Emergency Operations Center dashboard.



Aggregates signals from 4 data sources automatically



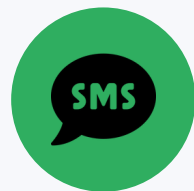
Fuses duplicate reports into consolidated incidents via record linkage



Scores every incident with a 5-factor hybrid triage algorithm



Displays a live GIS map with color-coded priority levels



Sends email dispatch alerts directly to field teams



Provides a real-time AI chatbot assistant for the EOC operator

What Makes PRATHAMIKTA Unique



Multi-Source Fusion

The only system merging 4 signal types into one incident automatically



Priority score

Every score shown with breakdown — never a black box



Instant email dispatch alerts

EOC operators can send instant email for dispatch alerts

Built for Real Emergency Operations

WHO USES THIS

- Nepal's NDRRMA (National Disaster Risk Reduction & Management Authority)
- District Emergency Operation Centers (DEOCs)
- Municipal Emergency Response Units

WHAT IT REPLACES

- | | | |
|--|---|--|
| Manual WhatsApp coordination | → | Automated priority queue |
| Duplicate incident tracking | → | Intelligent signal fusion |
| Subjective dispatch decisions | → | Algorithmic, transparent triage |
| Delayed field notification | → | Instant email dispatch alerts |



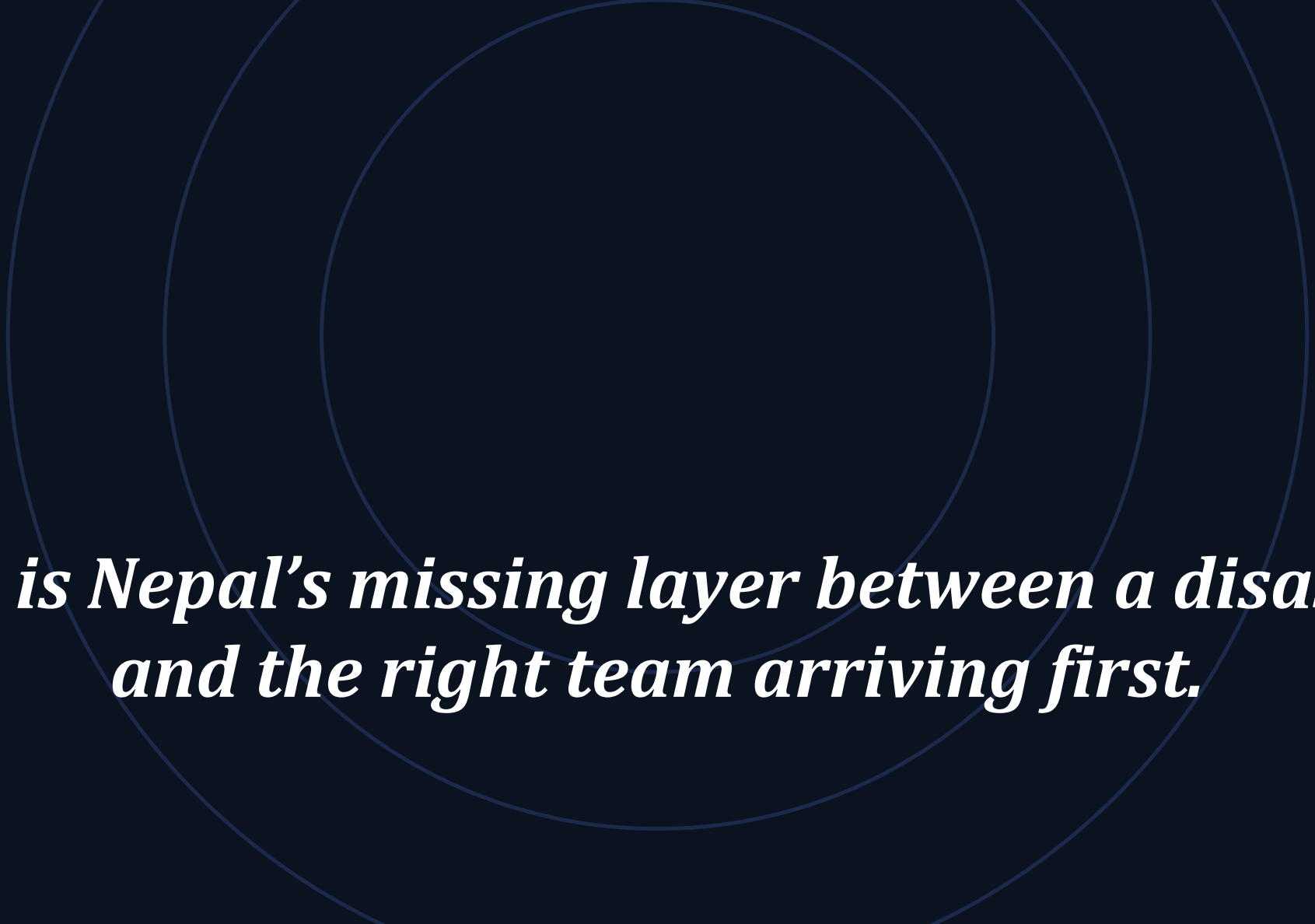
TIME SAVED

10 min

faster team dispatch can save lives in a real disaster scenario

15–30 minutes

currently spent by EOC operators manually sorting and cross-referencing reports — eliminated by PRATHAMIKA



PRATHAMIKA is Nepal's missing layer between a disaster happening and the right team arriving first.

"When 10 disasters strike at once, PRATHAMIKA tells you which one is screaming loudest — and why."

Team LumenX

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